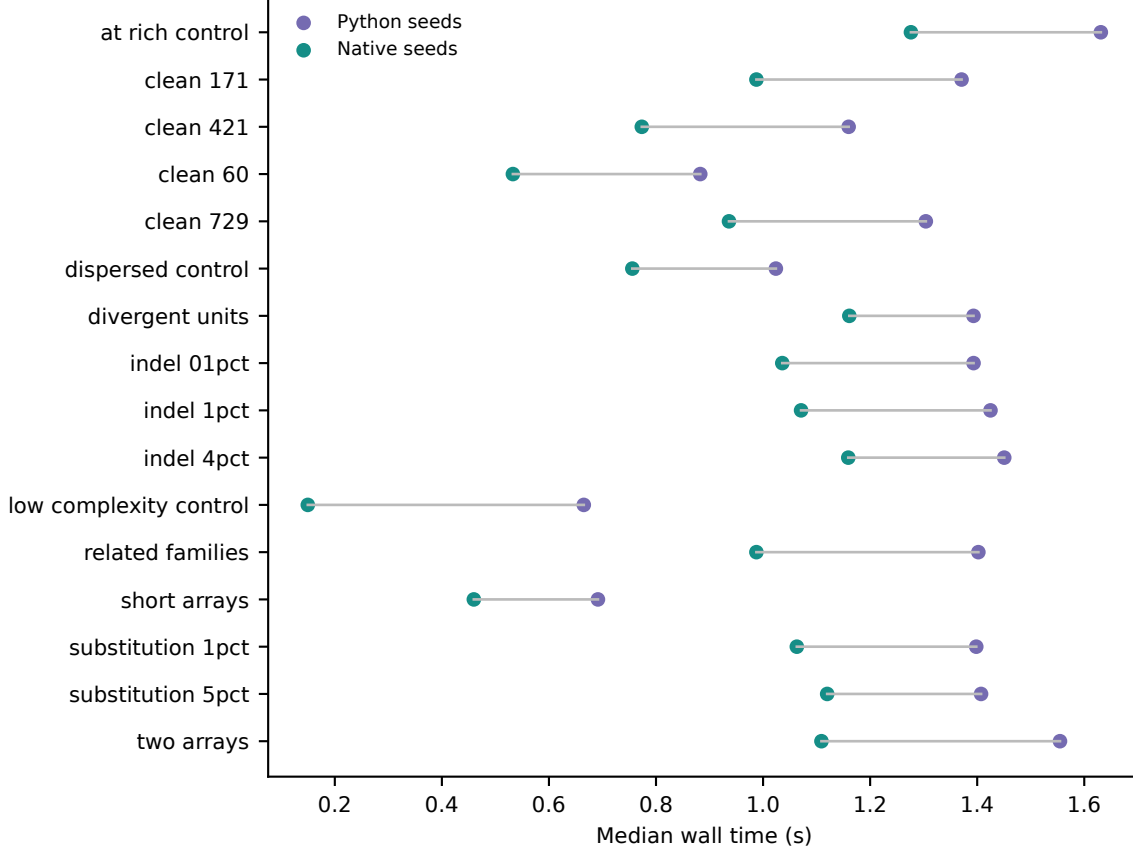
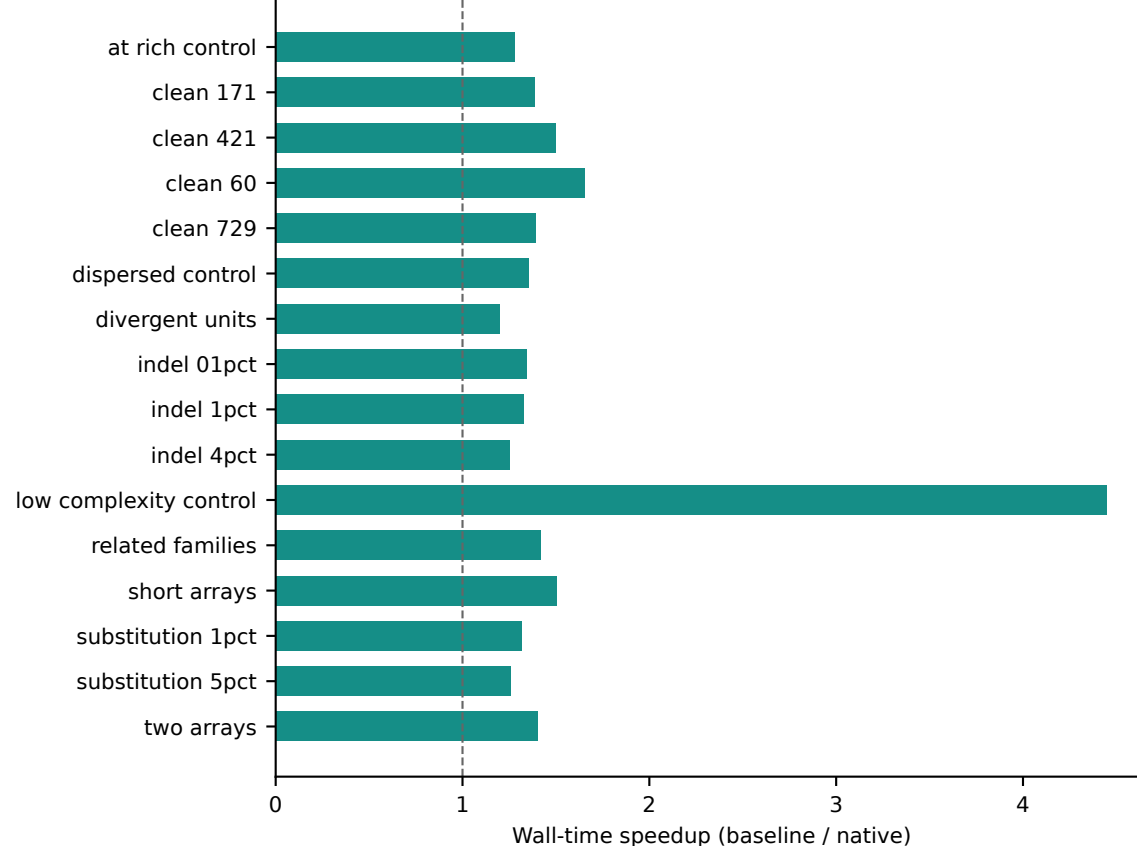


Native seed processing: paired development experiment

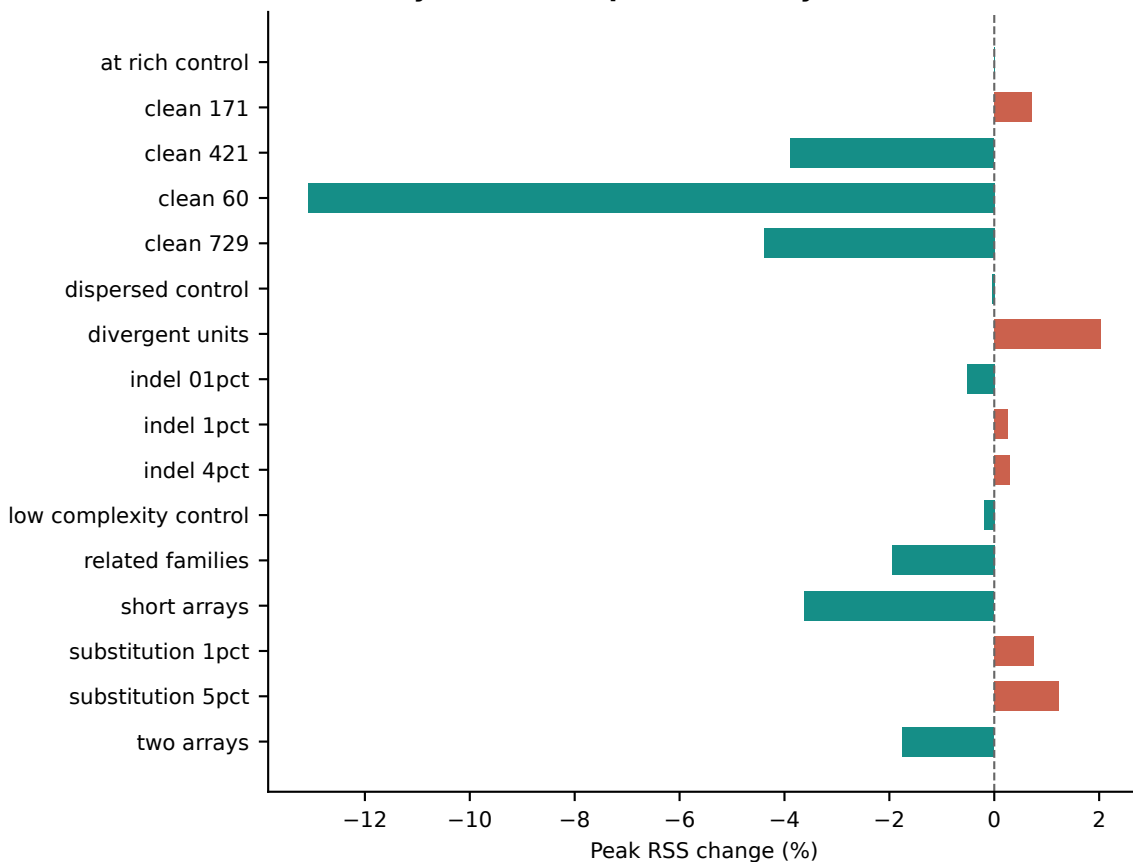
A Identical outputs, lower computation time



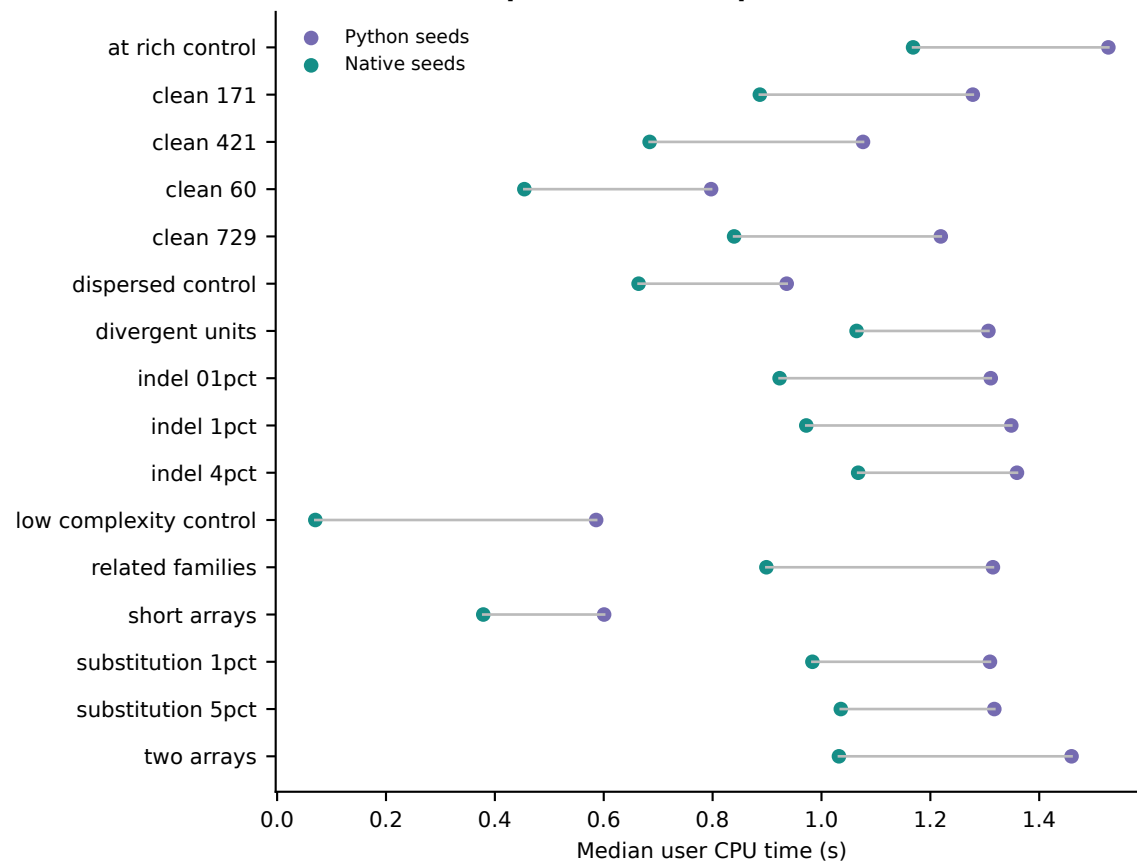
B Every scenario, including negative controls



C Memory does not improve in every scenario



D Identical outputs, lower computation time



16 datasets × 2 implementations × 3 repeats; 100 reads per dataset, seed 2101; six outputs byte-identical
Same-tool optimization only. Technical repeats are not biological replicates.